

COMP2501 Exercise 7

Due date: November 15, Friday 23:59, 2024

1. Following last week's exercise, in the `extdata` directory of the `dslabs` package, you will find a PDF file containing daily mortality data for Puerto Rico from Jan 1, 2015 to May 31, 2018. Please use the following code to load the trimmed data.

```
fn <- system.file("extdata",
"RD-Mortality-Report_2015-18-180531.pdf", package = "dslabs")
install.packages("pdftools")
library(pdftools)

txt <- pdf_text(fn)
page_nine <- txt[9]
library(stringr)
s <- str_split(page_nine, "\n")[[1]]
s <- str_trim(s)
```

For each question, please paste your code and result (if it's not provided).

1.1 We want to extract the numbers from the strings stored in `s`. However, there are many non-numeric characters that will get in the way. We can remove these, but before doing this we want to preserve the string with the column header, which includes the month abbreviation. Use the `str_which` function to find the rows with a header. Save these results to `header_index`. Hint: find the first string that matches the pattern `2015` using the `str_which` function.

1.2 Now we are going to define two objects: `month` will store the month and `header` will store the column names. Identify which row contains the header of the table. Save the content of the row into an object called `header`, then use `str_split` to help define the two objects we need. Hints: the separator here is one or more spaces. Also, consider using the `simplify` argument. You will see the result looks like this:

```
> header
[1] "2015" "2016" "2017" "2018"
> month
[1] "SEP"
```

1.3 Notice that towards the end of the page you see a *totals* row followed by rows with other summary statistics. Create an object called `tail_index` with the index of the *totals* entry.

1.4 Because our PDF page includes graphs with numbers, some of our rows have just one number (from the y-axis of the plot). Use the `str_count` function to create an object `n` with the number of numbers in each row. Hint: you can write a regex for number like this `\\d+`.

1.5 We are now ready to remove entries from rows that we know we don't need. The entry `header_index` and everything before it should be removed. Entries for which `n` is 1 should also be removed, and the entry `tail_index` and everything that comes after it should be removed as well.

1.6 Now we are ready to remove all the non-numeric entries. Do this using regex and the `str_remove_all` function. Hint: remember that in regex, using the upper case version of a special character usually means the opposite. So `\\D` means "not a digit". Remember you also want to keep spaces.

1.7 To convert the strings into a table, use the `str_split_fixed` function. Convert `s` into a data matrix with just the day and death count data. Hints: note that the separator is one or more spaces. Make the argument `n` a value that limits the number of columns to the values in the 4 columns and the last column captures all the extra stuff. Then keep only the first four columns.

1.8 Now you are almost ready to finish. Add column names to the matrix, including one called `day`. Also, add a column with the month. Call the resulting object `dat`. Finally, make sure the day is an integer not a character. Hint: use only the first five columns. Remove the last rows that don't match the formats. You will see the result looks like this:

```
> dat
```

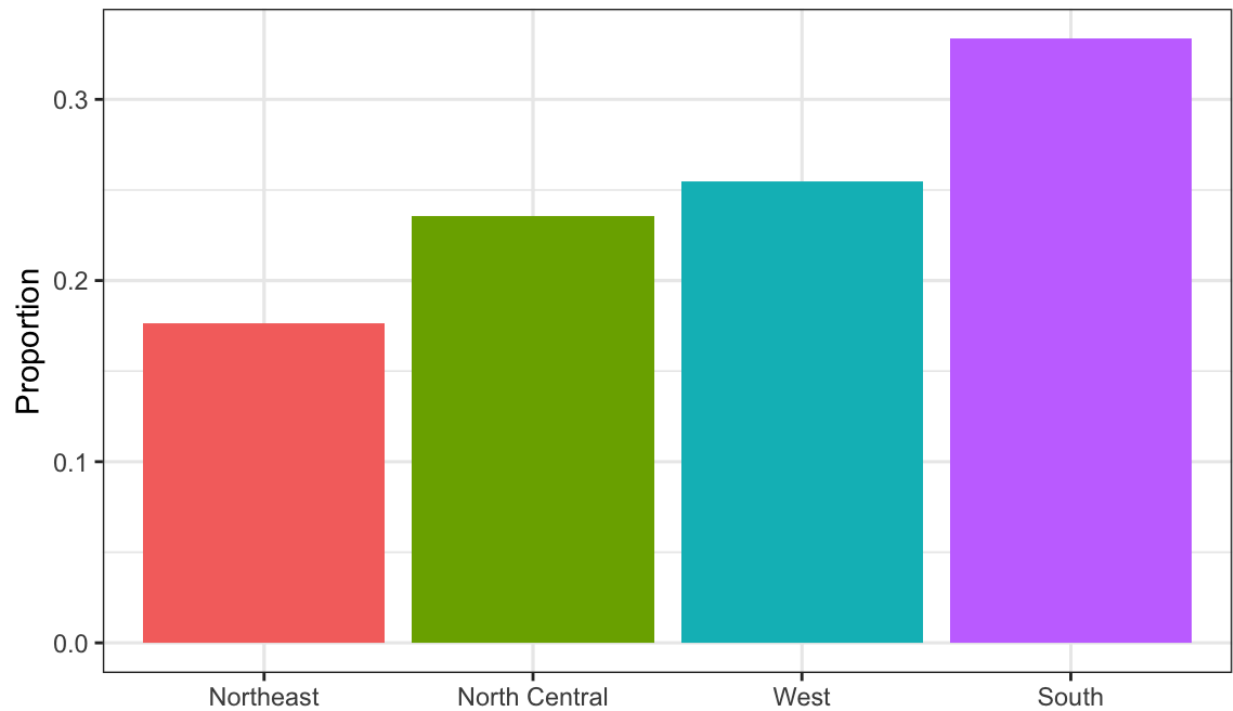
```
   day 2015 2016 2017 month
1    1   75   75   94   SEP
2    2   77   67   69   SEP
3    3   67   78   80   SEP
4    4   71   99   84   SEP
5    5   62   89   74   SEP
```

1.9 Now finish it up by tidying `tab` with the `pivot_longer` function. You will see the result looks like this:

```
> dat
```

```
# A tibble: 90 × 4
   day month  year deaths
<int> <chr> <int>   <int>
1     1  1 SEP    2015     75
2     1  1 SEP    2016     75
3     1  1 SEP    2017     94
4     2  2 SEP    2015     77
5     2  2 SEP    2016     67
```

2. In the `murders` dataset, the `region` is a categorical variable and the following is its distribution:

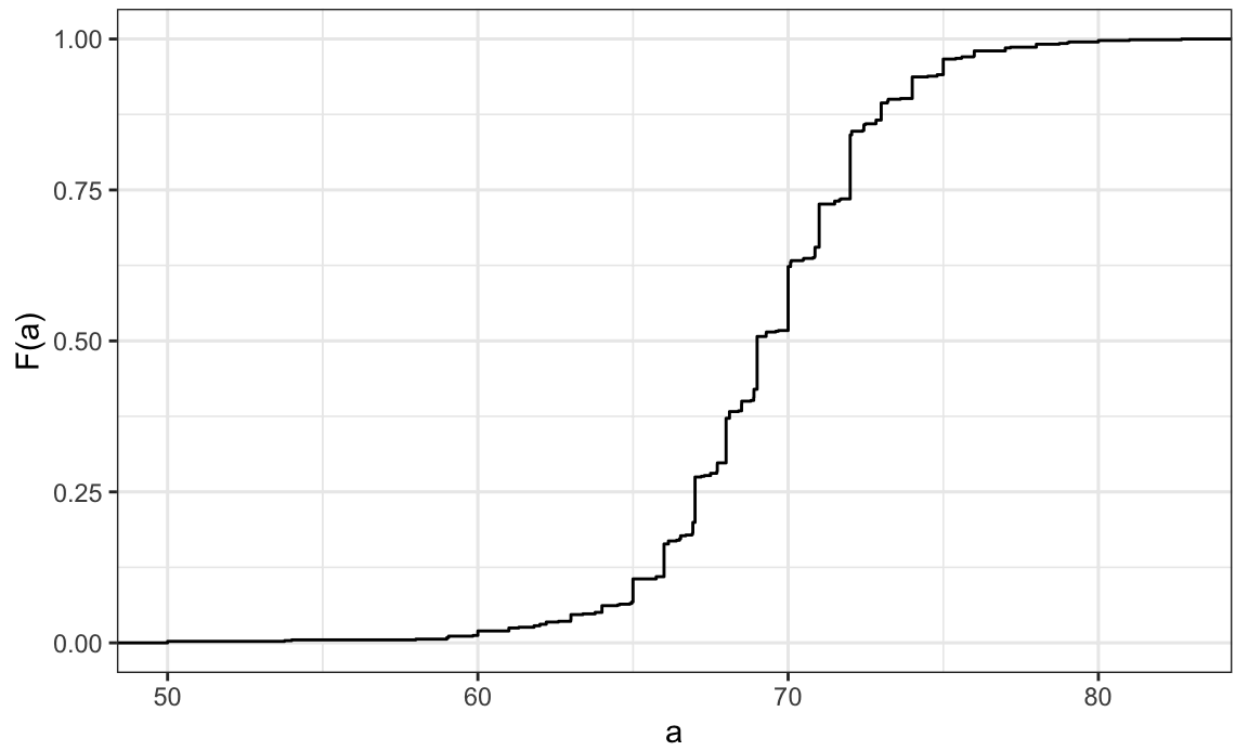


To the closest 5%, what proportion of the states are in the North Central region?

3. Which of the following is true:

- A. The graph above is a histogram.
- B. The graph above shows only four numbers with a bar plot.
- C. Categories are not numbers, so it does not make sense to graph the distribution.
- D. The colors, not the height of the bars, describe the distribution.

4. The plot below shows the eCDF for male heights:



Based on the plot, what percentage of males are shorter than 75 inches?

- A. 100%
- B. 95%
- C. 80%
- D. 72 inches